

Torque Specifications (Cont'd)

Part tightened	N-m	kgf-cm	ft-lbf
PS pump x Bracket	39	400	29
A/C compressor x Bracket	25	250	18
Suspension lower crossmember x Body	152	1,550	112

ENGINE MECHANICAL (3S-GTE)

Specifications

Idle speed				800 ± 50 rpm
Intake manifold vacuum	at idle speed			60 kPa 450 mmHg 17.7 in.Hg
Compression	at 250 rpm	STD Limit		1,128 kPa (11.5 kgf/cm ² , 164 psi) or more 883 kPa 9.0 kgf/cm ² 128 psi 98 kPa (1.0 kgf/cm ² , 14 psi) or less
	Difference of pressure between each cylinder			
Timing belt tensioner	Protrusion			8.5 – 9.5 mm 0.335 – 0.374 in.
Cylinder head	warpage	Cylinder block side Intake manifold side Exhaust manifold side	Limit Limit Limit	0.20 mm 0.0079 in. 0.20 mm 0.0079 in. 0.30 mm 0.0118 in.
	Valve seat	Refacing angle Contacting angle Contacting width		30°, 45°, 75° 45° 1.0 – 1.4 mm 0.039 – 0.055 in.
Valve guide bushing	Inside diameter			6.000 – 6.018 mm 0.2362 – 0.2369 in.
	Outside diameter (for repair part)	STD O/S 0.05		11.030 – 11.041 mm 0.4343 – 0.4347 in. 11.080 – 11.091 mm 0.4362 – 0.4367 in.
Valve	Valve overall length	STD	Intake Exhaust	100.50 mm 3.9567 in. 99.55 mm 3.9193 in.
		Limit	Intake Exhaust	99.80 mm 3.9291 in. 98.85 mm 3.8917 in.
	Valve face angle			44.5°
	Stem diameter		Intake Exhaust	5.960 – 5.975 mm 0.2346 – 0.2352 in. 5.955 – 5.970 mm 0.2344 – 0.2350 in.
	Stem oil clearance	STD	Intake Exhaust	0.025 – 0.058 mm 0.0010 – 0.0023 in. 0.030 – 0.063 mm 0.0012 – 0.0025 in.
		Limit	Intake Exhaust	0.08 mm 0.0031 in. 0.10 mm 0.0039 in.
	Margin thickness	STD Limit		0.8 – 1.2 mm 0.031 – 0.047 in. 0.5 mm 0.020 in.
Valve spring	Squareness	Limit		2.0 mm 0.079 in.
	Free length			44.43 mm 1.7492 in.
	Installed tension at 34.4 mm (1.354 in.)			201 – 236 N (20.5 – 24.1 kgf, 45.2 – 53.1 lbf)

Specifications (Cont'd)

Valve lifter	Lifter diameter		30.975 – 30.985 mm 1.2195 – 1.2199 in.
	Lifter bore diameter		31.000 – 31.021 mm 1.2205 – 1.2213 in.
	Oil clearance	STD	0.015 – 0.046 mm 0.0006 – 0.0018 in.
		Limit	0.07 mm 0.0028 in.
Manifold	Warpage	Limit	0.20 mm 0.0079 in.
Camshaft	Thrust clearance	STD	0.120 – 0.240 mm 0.0047 – 0.0094 in.
		Limit	0.30 mm 0.0118 in.
	Journal oil clearance	STD	0.025 – 0.062 mm 0.0010 – 0.0024 in.
		Limit	0.08 mm 0.0031 in.
	Journal diameter		26.959 – 26.975 mm 1.0614 – 1.0620 in.
	Circle runout	Limit	0.06 mm 0.0024 in.
	Cam lobe height	STD	41.010–41.110mm 1.6146–1.6185 in.
		Limit	41.090 – 41.190 mm 1.6177 – 1.6217 in.
T-VIS valve	Warpage	Limit	0.20 mm 0.0079 in.
Cylinder block	Cylinder head surface warpage	Limit	0.05 mm 0.0020 in.
	Cylinder bore diameter	Mark 1	86.000 – 86.010 mm 3.3858 – 3.3862 in.
		Mark 2	86.010 – 86.020 mm 3.3862 – 3.3866 in.
		Mark 3	86.020 – 86.030 mm 3.3866 – 3.3870 in.
		Limit	86.23 mm 3.3949 in.
Piston and piston ring	Piston diameter	Mark 1	85.920 – 85.930 mm 3.3827 – 3.3831 in.
		Mark 2	85.930 – 85.940 mm 3.3831 – 3.3835 in.
		Mark 3	85.940 – 85.950 mm 3.3835 – 3.3839 in.
	Piston oil clearance	STD	0.070 – 0.090 mm 0.0028 – 0.0035 in.
		Limit	0.110 mm 0.0043 in.
	Piston ring groove clearance	No.1	0.040 – 0.080 mm 0.0016 – 0.0031 in.
		No.2	0.030 – 0.070 mm 0.0012 – 0.0028 in.
	Piston ring end gap	STD	No.1 0.330 – 0.550 mm 0.0130 – 0.0217 in.
		No.2	0.450 – 0.670 mm 0.0177 – 0.0264 in.
		Oil	0.200 – 0.600 mm 0.0079 – 0.0236 in.
		Limit	No.1 0.85 mm 0.0335 in.
		No.2	0.97 mm 0.0382 in.
		Oil	0.90 mm 0.0354 in.
Connecting rod	Thrust clearance	STD	0.160 – 0.312 mm 0.0063 – 0.0123 in.
	Limit		0.35 mm 0.0138 in.
	Connecting rod bearing center wall thickness		
		STD	Mark 1 1.484 – 1.488 mm 0.0584 – 0.0586 in.
			Mark 2 1.488 – 1.492 mm 0.0586 – 0.0587 in.
			Mark 1.492 – 1.496 mm 0.0587 – 0.0589 in.
	Connecting rod oil clearance	STD	0.024 – 0.055 mm 0.0009 – 0.0022 in.
		Limit	0.023 – 0.069 mm 0.0009 – 0.0027 in.
			0.08 mm 0.0031 in.
	Rod bending Limit per 100 mm (3.94 in.)		0.05 mm 0.0020 in.
	Rod twist Limit per 100 mm (3.94 in.)		0.15 mm 0.0059 in.

Specifications (Cont'd)

Connecting rod (cont'd)	Bushing inside diameter		22.005 – 22.017 mm	0.8663 – 0.8668 in.
	Piston pin diameter		21.997 – 22.009 mm	0.8660 – 0.8665 in.
	Piston pin oil clearance	STD	0.005 – 0.011 mm	0.0002 – 0.0004 in.
		Limit	0.05 mm	0.0020 in.
Crankshaft	Thrust clearance	STD	0.020 – 0.220 mm	0.0008 – 0.0087 in.
		Limit	0.30 mm	0.0118 in.
	Thrust washer thickness		2.440 – 2.490 mm	0.0961 – 0.0980 in.
	Main journal oil clearance			
	STD	No-3 STD	0.025 – 0.044 mm	0.0010 – 0.0017 in.
		U/S 0.25	0.021 – 0.061 mm	0.0008 – 0.0024 in.
		Others STD	0.015 – 0.034 mm	0.0006 – 0.0013 in.
		U/S 0.25	0.029 – 0.069 mm	0.0011 – 0.0027 in.
	Limit		0.08 mm	0.0031 in.
	Main journal diameter	STD	54.988 – 55.003 mm	2.1653 – 2.1655 in.
		U/S 0.25	54.745 – 54.755 mm	2.1553 – 2.1557 in.
	Main bearing center wall thickness			
	STD No.3	Mark 1	1.992 – 1.995 mm	0.0784 – 0.0785 in.
		Mark 2	1.995 – 1.998 mm	0.0785 – 0.0787 in.
		Mark 3	1.998 – 2.001 mm	0.0787 – 0.0788 in.
		Mark 4	2.001 – 2.004 mm	0.0788 – 0.0789 in.
		Mark 5	2.004 – 2.007 mm	0.0789 – 0.0790 in.
	Others	Mark 1	1.997 – 2.000 mm	0.0786 – 0.0787 in.
		Mark 2	2.000 – 2.003 mm	0.0787 – 0.0789 in.
		Mark 3	2.003 – 2.006 mm	0.0789 – 0.0790 in.
		Mark 4	2.006 – 2.009 mm	0.0790 – 0.0791 in.
		Mark 5	2.009 – 2.012 mm	0.0791 – 0.0792 in.
	Crank pin diameter	STD	47.985 – 48.000 mm	1.8892 – 1.8898 in.
		U/S 0.25	47.745 – 47.755 mm	1.8797 – 1.8801 in.
		Limit	0.06 mm	0.0024 in.
	Circle runout	Limit	0.02 mm	0.0008 in.
	Main journal taper and out-of-round	Limit	0.02 mm	0.0008 in.
	Crank pin taper and out-of-round			

Torque Specifications

Part tightened		N·m	kgf-cm	ft-lbf
Oil pump pulley x Oil pump drive shaft		35	355	26
No.2 idler pulley x Cylinder block		43	440	32
No.1 idler pulley bracket x Cylinder head		52	530	38
Crankshaft pulley x Crankshaft		108	1,100	80
Camshaft timing pulley x Camshaft		59	600	43
	For SST	41	420	30
Timing belt tensioner x Cylinder head		21	210	15
Cylinder head x Cylinder block	1st	49	500	36
	2nd	Turn 90°		
Camshaft bearing cap x Cylinder head		19	190	14
No.3 timing belt cover x Cylinder head		8.8	90	78 in.-lbf

Torque Specifications (Cont'd)

Part tightened		N·m	kgf-cm	ft-lbf
Cylinder head cover x Cylinder head		2.5	25	21 in-lbf
Intake manifold x Cylinder head		19	195	14
Intake manifold stay x Intake manifold		25	260	19
Intake manifold stay x Cylinder block		25	260	19
Water by-pass pipe x Water pump cover		7.8	80	69 in-lbf
Water outlet x Cylinder head		39	400	29
EGR valve x Intake manifold		19	195	14
EGR pipe x Cylinder head		25	260	19
LH engine hanger x Cylinder head	12 mm head bolt	19	195	14
	14 mm head bolt	39	400	29
Exhaust manifold x Cylinder head		52	530	38
Catalytic converter x Turbine outlet elbow		29	300	22
Catalytic converter stay x Catalytic converter		59	600	43
No.1 alternator bracket x Cylinder head		39	400	29
Main bearing cap x Cylinder block		59	600	43
Connecting rod cap x Connecting rod		67	680	49
Rear oil seal retainer x Cylinder block		9.3	95	82 in-lbf
Knock sensor x Cylinder block		44	450	33
RH engine mounting bracket x Cylinder block		52	530	38
PS pump bracket x Cylinder block		43	440	32
Rear end plate x Cylinder block		9.3	95	82 in-lbf
Flywheel x Crankshaft		108	1,100	80
LH engine mounting bracket x Transaxle		52	530	38
LH engine mounting insulator x LH mounting bracket		63	650	47
LH engine mounting insulator x Body		87	890	64
RH engine mounting insulator x RH mounting bracket		52	530	38
RH engine mounting insulator x Body		87	890	64
RH engine mounting stay x RH mounting insulator		73	740	54
RH engine mounting stay x No.1 alternator bracket		73	740	54
LH engine mounting stay x LH mounting insulator		21	210	15
LH engine mounting stay x Transaxle		21	210	15
Front engine mounting bracket x Transaxle		77	790	57
Rear engine mounting bracket x Transaxle		77	790	57
Engine mounting center member x Body		52	530	38
Engine mounting center member x Front mounting insulator		73	740	54
Engine mounting center member x Rear mounting insulator		73	740	54
Front engine mounting insulator x Front mounting insulator		87	890	64
Rear engine mounting insulator x Rear mounting insulator		87	890	64
PS pump x PS pump bracket	Adjusting bolt	39	400	29
	Others	43	440	32
A/C compressor x Cylinder block		27	280	20
Suspension lower crossmember x Body		152	1,550	112
Transaxle oil cooler tube x Oil cooler hose		34	350	25
Suspension upper brace x Body	Bolt	21	210	15
	Nut	64	650	47